

STRATHMORE UNIVERSITY
BSC. COMPUTER NETWORKS AND CYBER SECURITY

CNS 3106: WIRELESS COMMUNICATIONS

Assignment No. 1

Date Due: **4th May 2026**

(20 Marks)

1. A wireless internet provider is experiencing poor signal coverage inside buildings in the outskirts of Nairobi, even though the outdoor signal strength is strong. The system operates at 2.4 GHz, and base stations are already installed at optimal outdoor locations. As a communication systems engineer, propose and justify a solution to improve indoor signal coverage using principles of Radio Wave Propagation and transmission systems. Your answer should consider: (10 Marks)
 - a. The causes of signal degradation when radio waves enter buildings
 - b. At least two different technical solutions (e.g., antenna design, frequency choice, repeaters, or system redesign)
 - c. Trade-offs between cost, complexity, and performance
 - d. How your solution affects propagation characteristics such as penetration, reflection, and attenuation
2. A telecommunications company plans to establish a 35 km wireless communication link between two rural towns operating at 900 MHz. The environment consists of open land with scattered vegetation and small hills. The maximum available transmit power is 5 W, and the system must achieve a minimum received power of -95 dBm for reliable operation. Design a complete communication link for this scenario using appropriate principles of Radio Wave Propagation. Your answer should include:
 - a. Selection and justification of a suitable propagation model
 - b. A detailed link budget (including assumptions made)
 - c. Choice of antenna types, gains, and placement (heights)
 - d. Consideration of real-world factors affecting signal propagation

Guidelines

- This is group work (maximum of 4 members per group).
- One group member should submit the work via the eLearning platform on behalf of the group.
- All group members must actively participate, and their names and registration numbers should be included on the cover page.
- Submissions should reflect a clear understanding of the concepts. The use of AI-generated content without proper understanding or modification is discouraged and may be subject to review and possibility of your submission being cancelled.